Paper Stack

(CV, NLP, LLM, VLM, Diffusion, Agentic AI, RAG)

Prerequisite: You should have a basic understanding of deep learning before reading this paper. I've created the "Before LLMs" series, which brings together some of the best free resources for learning deep learning fundamentals.

I highly recommend going through these materials before starting the paper. At a minimum, complete the first 4 weeks of the series so you have the foundational knowledge needed to understand the concepts in the paper.

Week1(Computer Vision)

1. **AlexNet:** https://proceedings.neurips.cc/paper_files/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf?utm_source=Securitylab.ru
2. **VGGNet:** https://arxiv.org/abs/1409.1556
3. **ResNet:** https://arxiv.org/abs/1512.03385

Week2(NLP)

1. **Word2Vec:** https://arxiv.org/abs/1301.3781
2. **Seq2Seq:**https://arxiv.org/pdf/1409.3215
3. **Attention:** https://arxiv.org/pdf/1409.0473

Week3(NLP)

1. **Attention is all you need:** https://arxiv.org/pdf/1706.03762
2. **BERT:** https://arxiv.org/abs/1810.04805
3. **GPT 1:** https://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf

Week4(CV)

1. **ViT:** https://arxiv.org/pdf/2010.11929
2. **Swin Transformer:** https://arxiv.org/abs/2103.14030
3. **PVT:** https://arxiv.org/pdf/2102.12122

Week5

1. **GPT-2:** https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf
2. **GPT-3:** https://arxiv.org/abs/2005.14165
3. **InstructGPT:** https://arxiv.org/abs/2005.14165

Week6

Coming Soon….